

EXP 18 :

Aim

To write a Pandas program to split a DataFrame into groups based on **school code** and **class** using the `groupby()` function.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame with student details.
3. Display the original DataFrame.
4. Use the `groupby()` method with the columns **school** and **class**.
5. Iterate through each group.
6. Display the group name and corresponding records.

Code

```
import pandas as pd

data = {
    'school': ['s001', 's002', 's003', 's001', 's002', 's004'],
    'class': ['V', 'V', 'VI', 'VI', 'V', 'VI'],
    'name': ['Alberto Franco', 'Gino Mcneil', 'Ryan Parkes',
            'Eesha Hinton', 'Gino Mcneil', 'David Parkes'],
    'date_of_Birth': ['15/05/2002', '17/05/2002', '16/02/1999',
                    '25/09/1998', '11/05/2002', '15/09/1997'],
    'age': [12, 12, 13, 13, 14, 12],
    'height': [173, 192, 186, 167, 151, 159],
    'weight': [35, 32, 33, 30, 31, 32],
    'address': ['street1', 'street2', 'street3',
               'street1', 'street2', 'street4']
}

df = pd.DataFrame(data)

print("Original DataFrame:\n")
print(df)

print("\nGrouped Data:\n")

grouped = df.groupby(['school', 'class'])

for group_name, group_data in grouped:
    print("Group:", group_name)
    print(group_data)
    print("-" * 50)
```

Output

Original DataFrame:

	school	class	name	date_Of_Birth	age	height	weight	address
0	s001	V	Alberto Franco	15/05/2002	12	173	35	street1
1	s002	V	Gino Mcneil	17/05/2002	12	192	32	street2
2	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
3	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
4	s002	V	Gino Mcneil	11/05/2002	14	151	31	street2
5	s004	VI	David Parkes	15/09/1997	12	159	32	street4

Grouped Data:

Group: ('s001', 'V')

	school	class	name	date_Of_Birth	age	height	weight	address
0	s001	V	Alberto Franco	15/05/2002	12	173	35	street1

Group: ('s001', 'VI')

	school	class	name	date_Of_Birth	age	height	weight	address
3	s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1

Group: ('s002', 'V')

	school	class	name	date_Of_Birth	age	height	weight	address
1	s002	V	Gino Mcneil	17/05/2002	12	192	32	street2
4	s002	V	Gino Mcneil	11/05/2002	14	151	31	street2

Group: ('s003', 'VI')

	school	class	name	date_Of_Birth	age	height	weight	address
2	s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3

Group: ('s004', 'VI')

	school	class	name	date_Of_Birth	age	height	weight	address
5	s004	VI	David Parkes	15/09/1997	12	159	32	street4

Result

The DataFrame was successfully **split into groups based on the school and class columns** using the Pandas `groupby()` function. Each group contains only the records that belong to the same school code and class.

EXP - 19

Aim

To write a Pandas program to display the dimensions (shape) of the World Alcohol Consumption dataset and extract the column names.

Algorithm

1. Import the Pandas library.
2. Load the World Alcohol Consumption dataset into a DataFrame.
3. Display the shape of the DataFrame using the shape attribute.
4. Display the column names using the columns attribute.
5. Print the results.

Code

```
import pandas as pd

# Load the dataset
df = pd.read_csv("world_alcohol.csv")

# Display the shape of the dataset
print("Shape of the dataset:")
print(df.shape)

# Display the column names
print("\nColumn Names:")
print(df.columns)
|
```

Output

```
Shape of the dataset:  
(10, 5)
```

```
Column Names:  
Index(['Year', 'WHO region', 'Country', 'Beverage Types', 'Display Value'], dtype='str')  
|
```

Result

The dimensions (shape) of the World Alcohol Consumption dataset and the column names were successfully displayed using the shape and columns attributes in Pandas.

EXP 20:

Aim

To write a Pandas program to find the index of rows containing a given substring in a DataFrame column.

Algorithm

1. Import the Pandas library.
2. Load the dataset into a DataFrame.
3. Use the str.contains() method to search for the given substring in the required column.
4. Retrieve the matching row indices using the index attribute.
5. Display the indices.

Code

```
import pandas as pd

# Load the dataset
df = pd.read_csv("world_alcohol.csv")

# Find the index of rows containing the substring "Wine"
result = df[df["Beverage Types"].str.contains("Wine", case=False, na=False)]

print("Index of matching rows:")
print(result.index.tolist())
```

Output

```
Index of matching rows:
[0, 2, 5, 8]
```

Result

The indices of the rows containing the specified substring were successfully found using the str.contains() method and displayed using the index attribute in Pandas.